Smart Grid, Smarter Habits

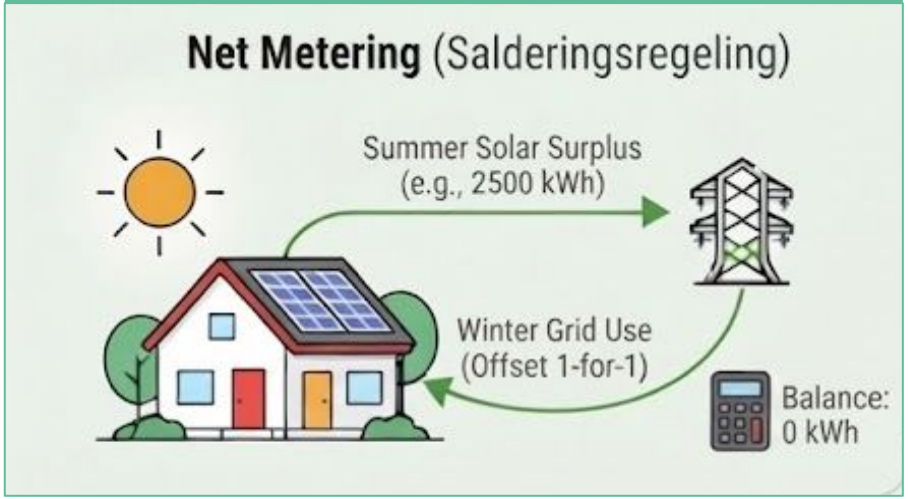
Optimizing Household Energy Spend Under 2027 Dutch Solar Energy Market Rules

By Amanda Joseph



The Dutch Energy Landscape: The Era of Free Energy Banking

The Past / Current Rule
(until 31 December 2026)



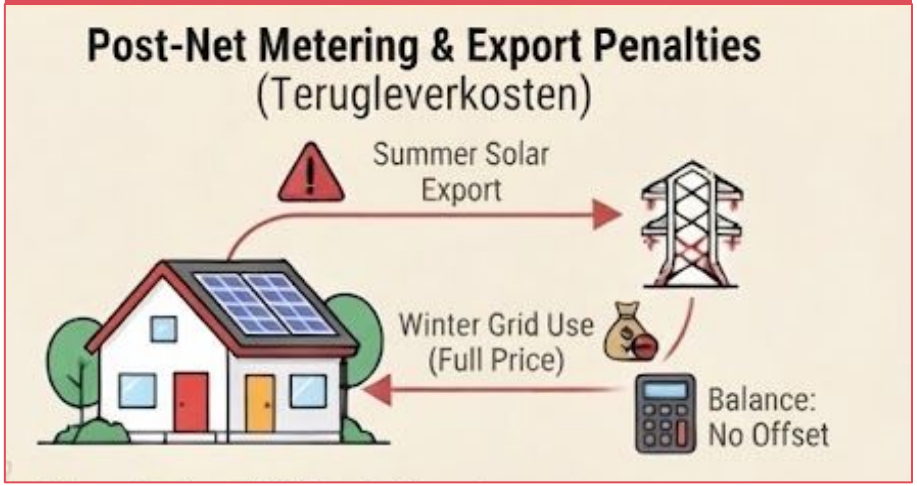
1-for-1 Offset
Summer export directly cancels out winter consumption.

Zero Extra Fees
No supply charges, network costs, or taxes on balanced kWh.

Financial Shield
Solar injection acts as a literal risk-free energy bank.

The Dutch Energy Landscape: The End of Net Metering

The New Reality
(from 1 January 2027)



No More 1-for-1 Offset
Surplus energy can no longer be directly subtracted from later use.

Export Penalties
Suppliers charge strict *terugleverkosten* for feeding back into the grid.

Self-Consumption King
Using your own solar power directly becomes essential to avoid penalties.

Building the Dataset: Weather, Home & Market

KNMI

Royal Netherlands Meteorological Institute

ENTSO-E API

European Network of Transmission System Operators for Electricity Transparency Platform

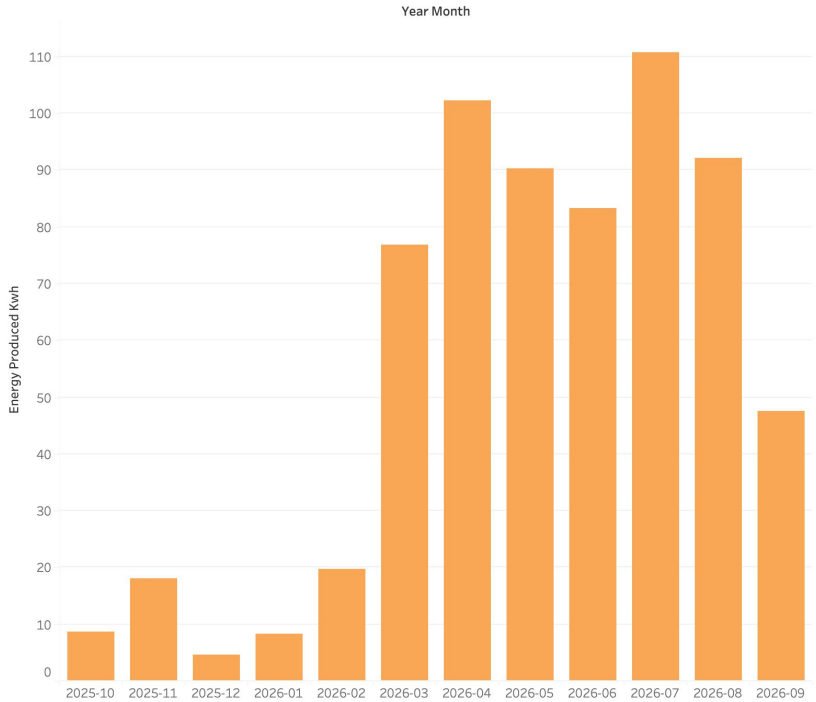


HOME ASSISTANT

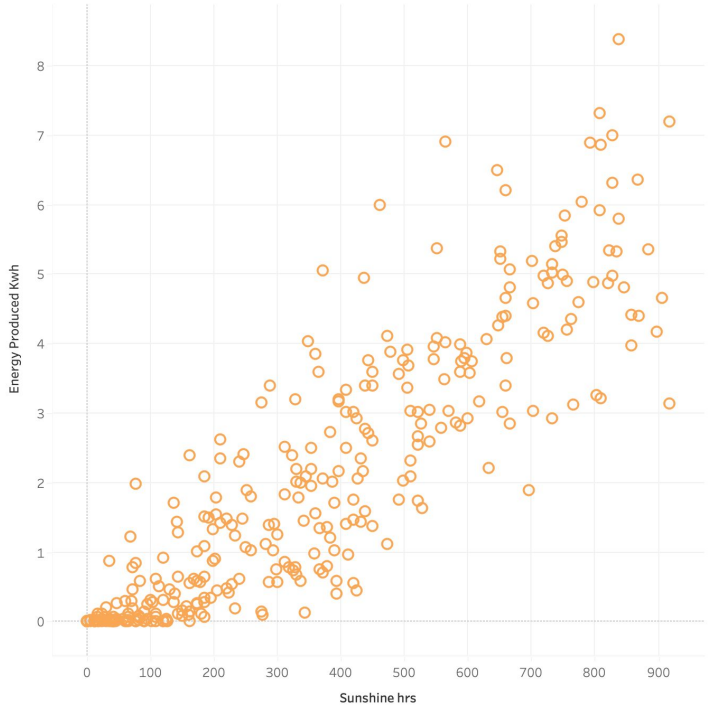
Source: P1 / Solar export - High frequency (1-minute intervals)

The Weather Factor: Dutch Solar Reality

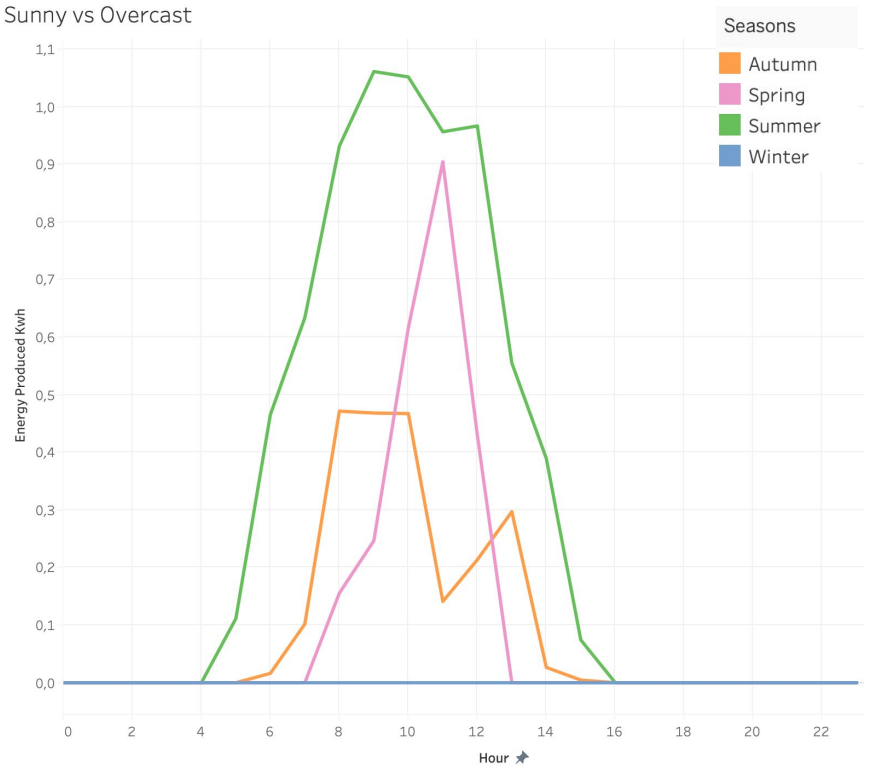
Solar Yield

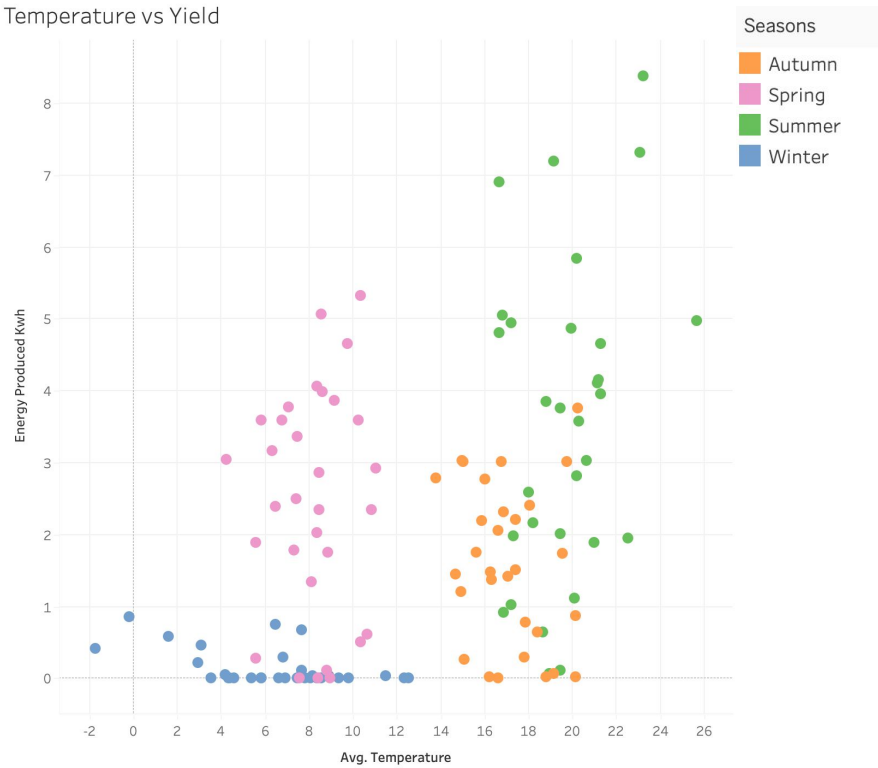


Sunshine vs Solar



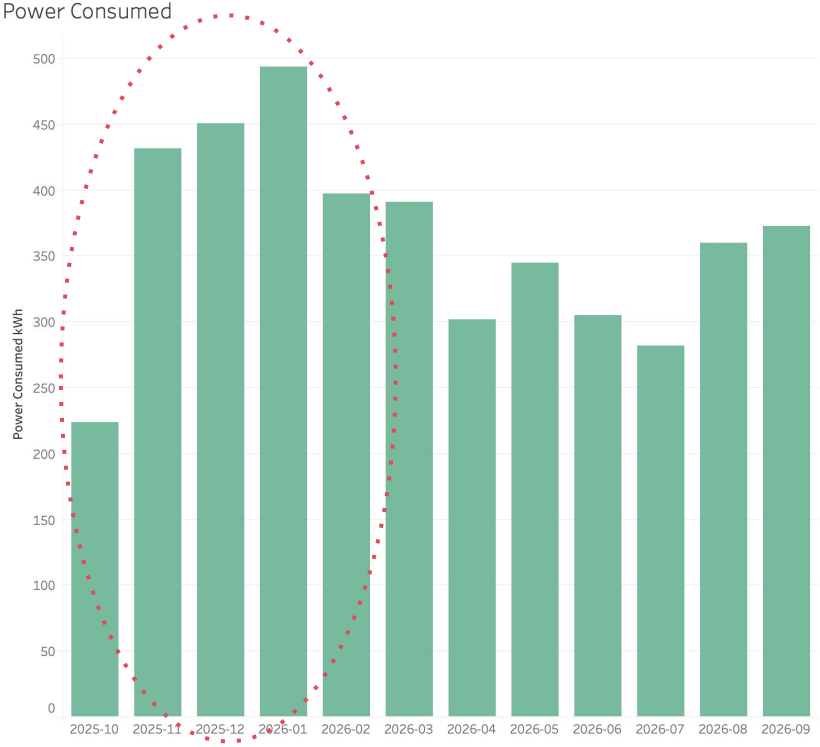
How Temperature and Daily Volatility Shape Yield



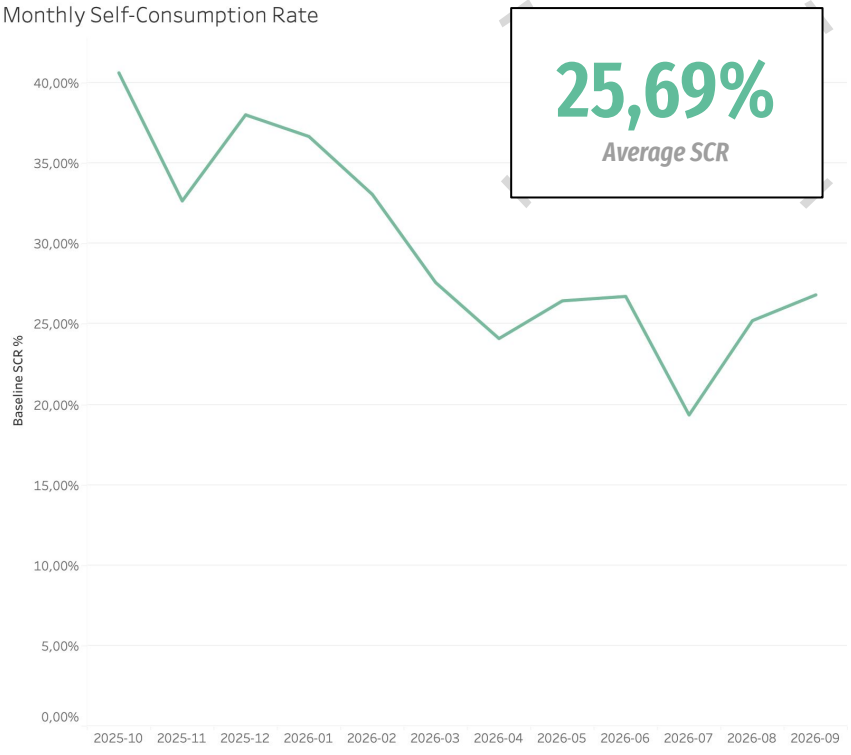


Household Demand and Self-Consumption

Power Consumed



Monthly Self-Consumption Rate

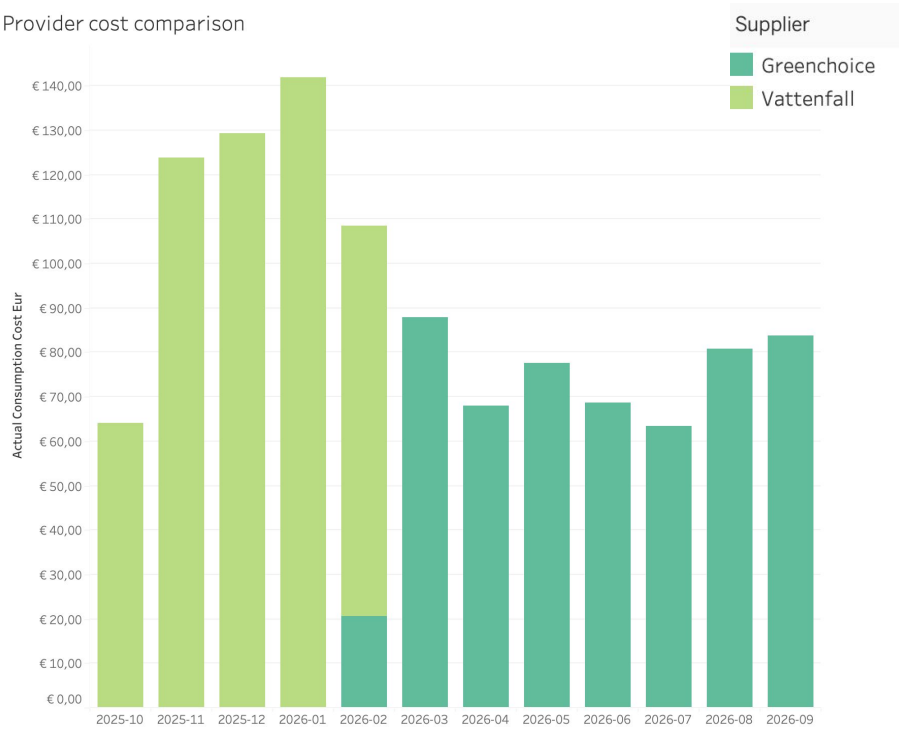


25,69%

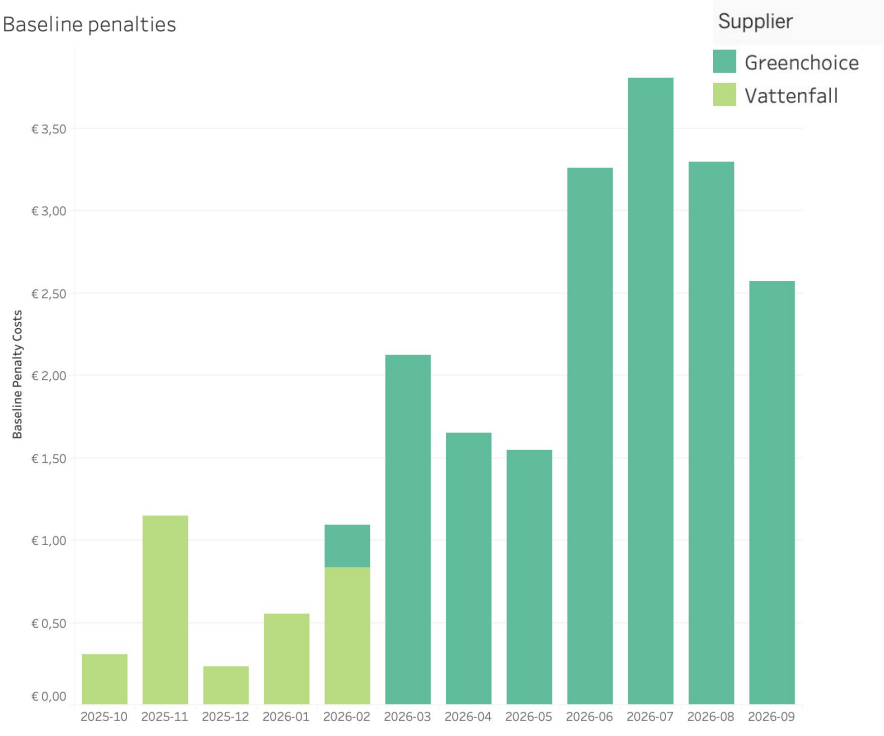
Average SCR

Household Demand and Self-Consumption

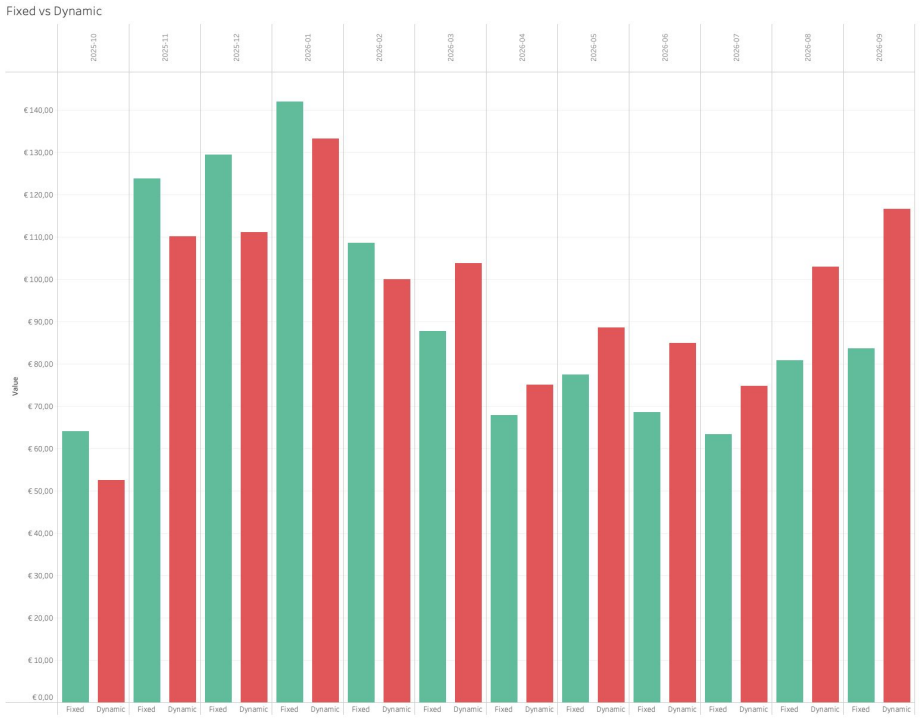
Provider cost comparison



Baseline penalties



Fixed vs. Dynamic: Annual Cost Comparison



Total Fixed Contract Cost

€1081

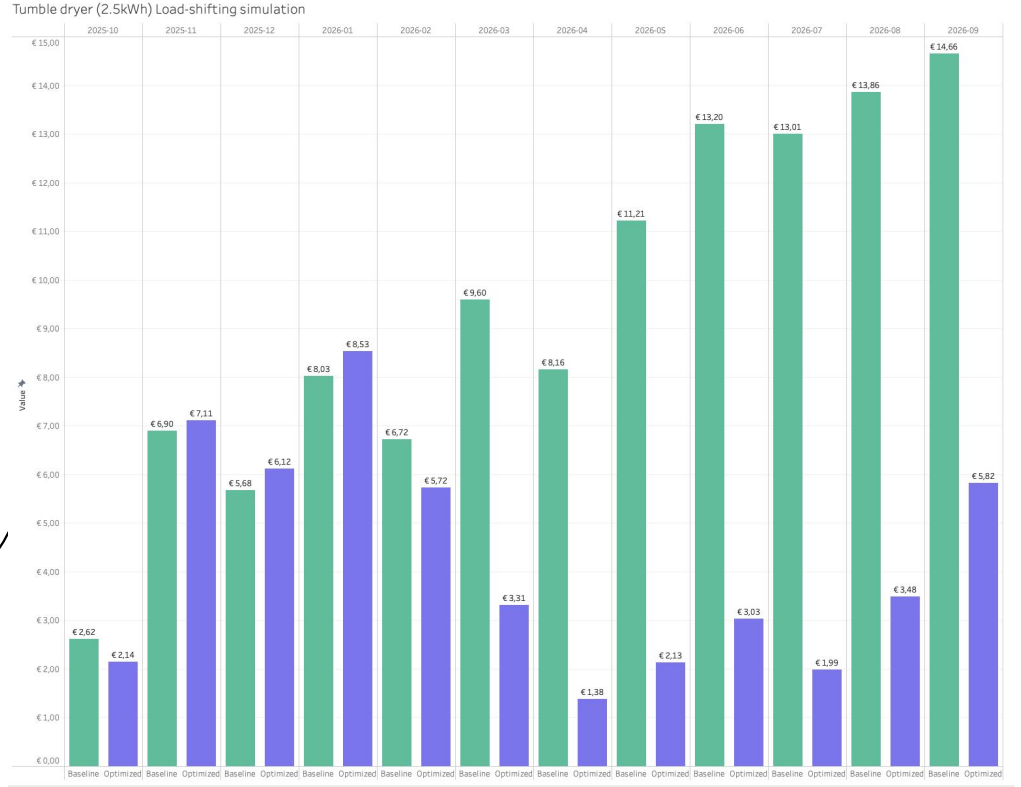
Total Dynamic Contract Cost

€1142

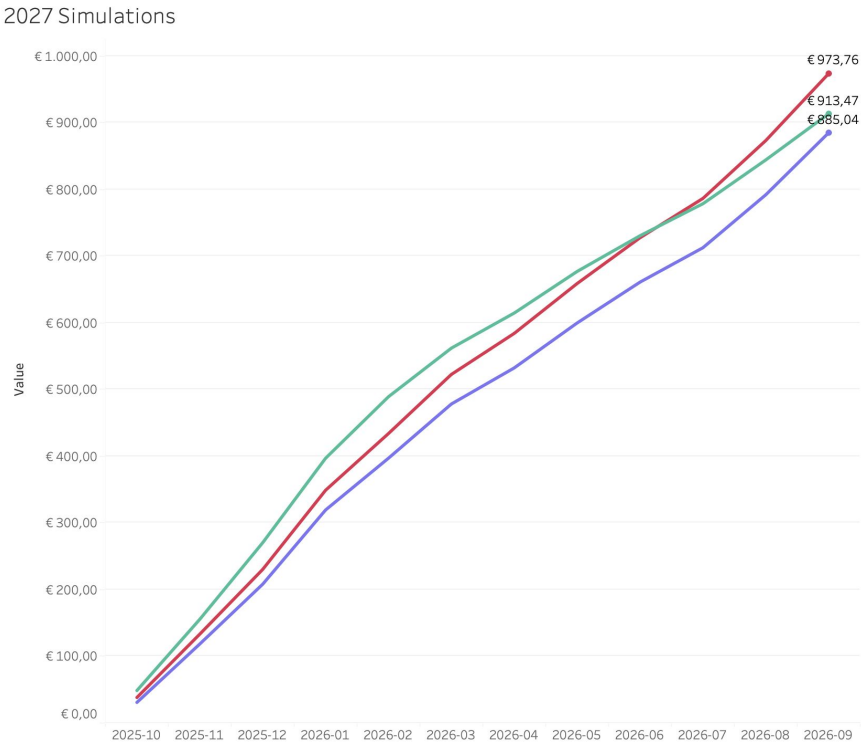
Load-Shifting Simulation: The Power of Midday Appliance Scheduling

Total Annual Savings
€110,25





2027 Outlook: The Cumulative Value of Smart Optimization



Dynamic Contract
€973,76

Fixed Contract
€913,47

Optimized Dynamic Contract
€885,04

Final Takeaway

01

Smart Scheduling

Smart midday load-shifting could save you over €110+ per year without needing a costly home battery

02

Avoid the Passive Dynamic Trap

A dynamic contract *without* automation is the most expensive path (€973.76).

03

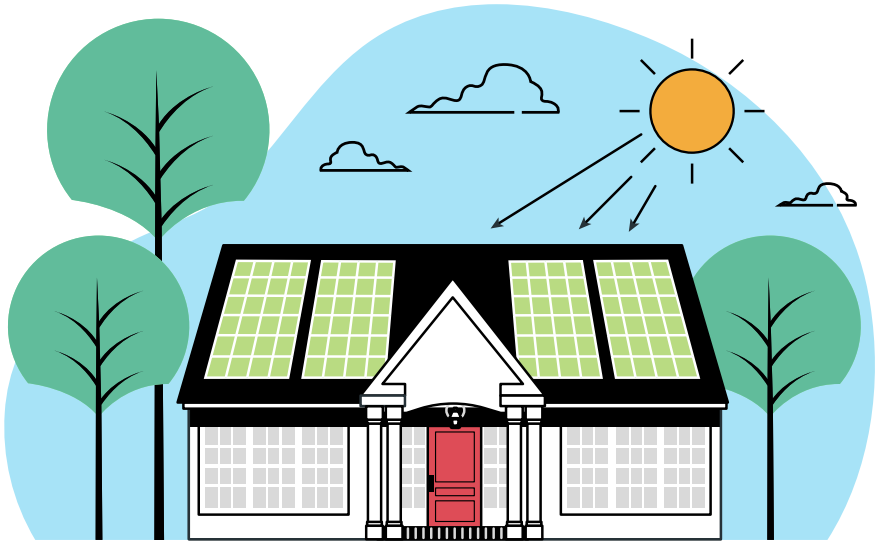
Align with the Sun

Deep winter yields no solar benefits, but spring and summer midday optimization more than makes up for it.

04

The Bottom Line

By aligning our daily routines with EPEX spot-price lows, we can reduce the penalty costs and take control of our energy bills post-2027 on our own terms.



Thank You!